

UML Fundamentals

Update Information for Sparx Systems Enterprise Architect Version 15.

Sparx Systems has changed the user interface for Enterprise Architect (EA) over the years, and there are now some clear differences in the way that it works compared to the version that was current when the course was created. Although the course tries just to use EA as an example of a modelling tool rather than being a course on EA, the differences that have been introduced can be an obstacle to getting started with modelling using the software, so these notes are intended to clarify some of the differences and help you get started with EA based on Version 15.

The notes are organised with sections for the first six chapters in the course where there are relevant changes. At some points, I have included the timing in minutes and seconds in brackets like this — (1:53) — to indicate the stage in the video that a particular note applies to. Names of menus, ribbons, clickable icons, dialogues and other screen elements are highlighted in bold in the text below. We are working on the assumption that once you have got through the first six chapters, you will have found out enough about the differences that have been made to make sense of the user interface and find your way around it for the rest of the course. Six chapters means that examples of both structural and behavioural diagrams have been covered.

1. Introduction

1.1 About the Course

Although it doesn't say so, you can just draw the diagrams on paper if you don't have access to a modelling tool. The advantage of the modelling tool is that it holds all the associated data about the model in its database.

1.2 About the Author

No updates.

1.3 About UML

The current version of UML is 2.5.1, which was finalised in December 2017. You can download the specification as a PDF from the OMG website at:

<https://www.omg.org/spec/UML>

The OCUP Certification now covers version 2.5 of UML at all three levels, which it didn't when the course was written. Information about certification is available on the OMG website at:

<https://www.omg.org/ocup-2/index.htm>

The ISO/IEC Standard is still based on the version from 2012.

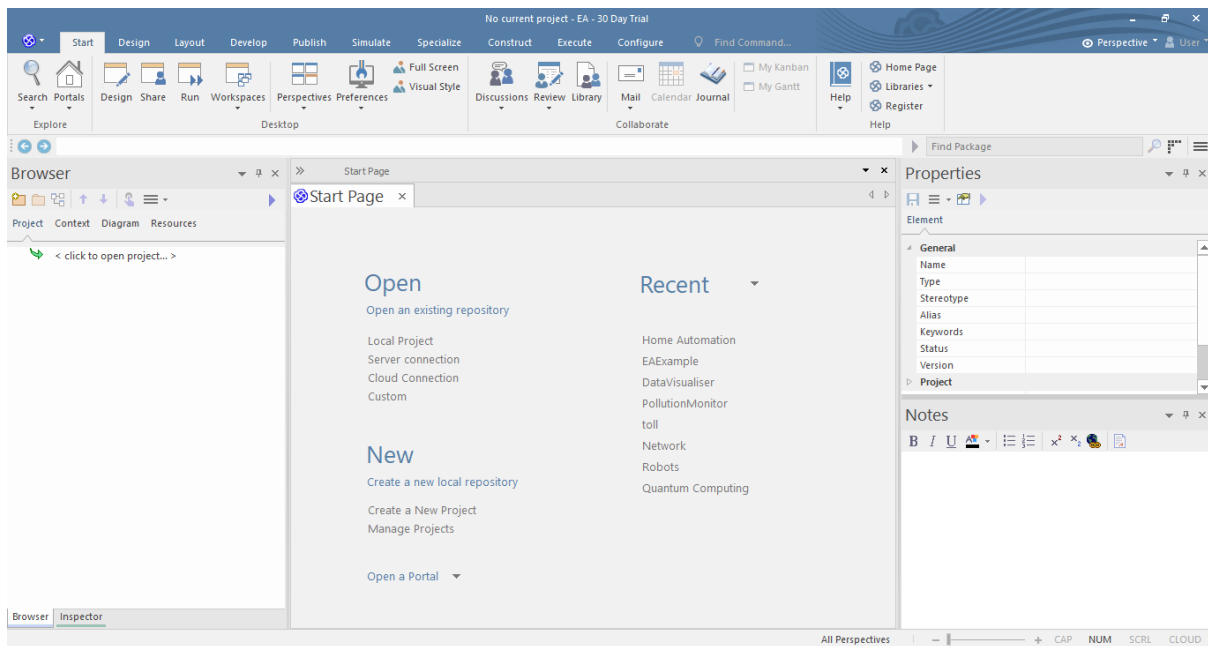
2. Diagrams in UML

2.1 Models and Diagrams

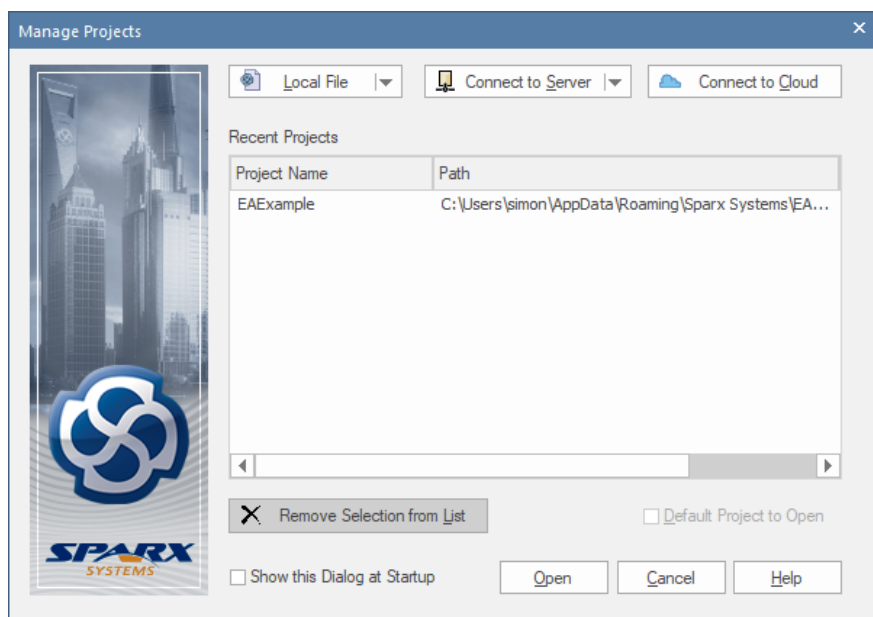
No updates.

2.2 Introduction to the Modelling Software

(0:39) When you start up EA 15, the window is likely to look like the following diagram. EA now uses ribbons in line with current Microsoft user interface style. At the time the course was made, EA used menus and toolbars. You can still enable toolbars, and there's a dialogue under **Start > Preferences > Toolbars...** that allows you to configure floating toolbars. These floating toolbars can be docked at the top of the window under the ribbon if you want.

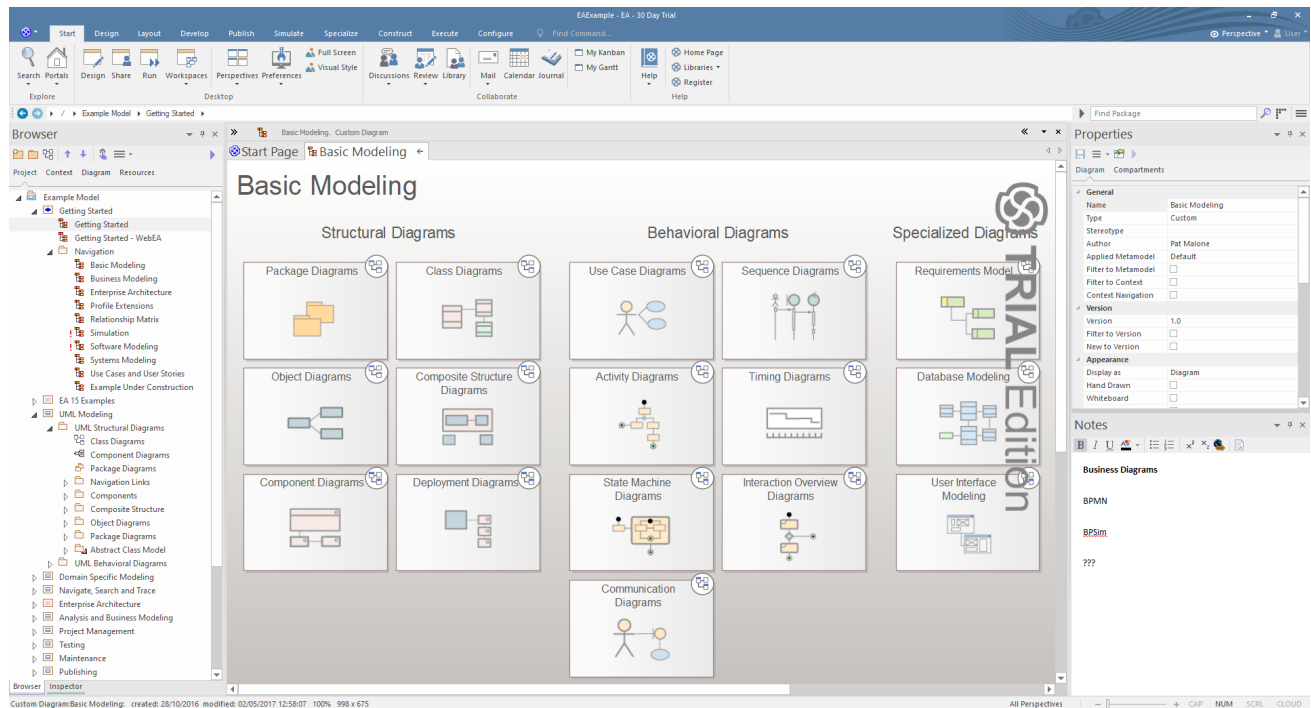


It displays the **Start Page** in the **Diagram Pane**. The **Manage Projects** dialogue shown in the video may not be displayed. If you want to, you can open it through the menu using the left-hand menu that has the Sparx Systems EA symbol in it, which I'll refer to as the **File** menu. The option is **File > Open Project...** It opens the following dialogue box.



You can then select the project you want from the list. However, you can open the **EAExample** project by just clicking on it in the **Start Page**, where it appears under **Recent**. If you want to open a project that you haven't opened previously or create a new project, you can click on the **Local File** button at the top left.

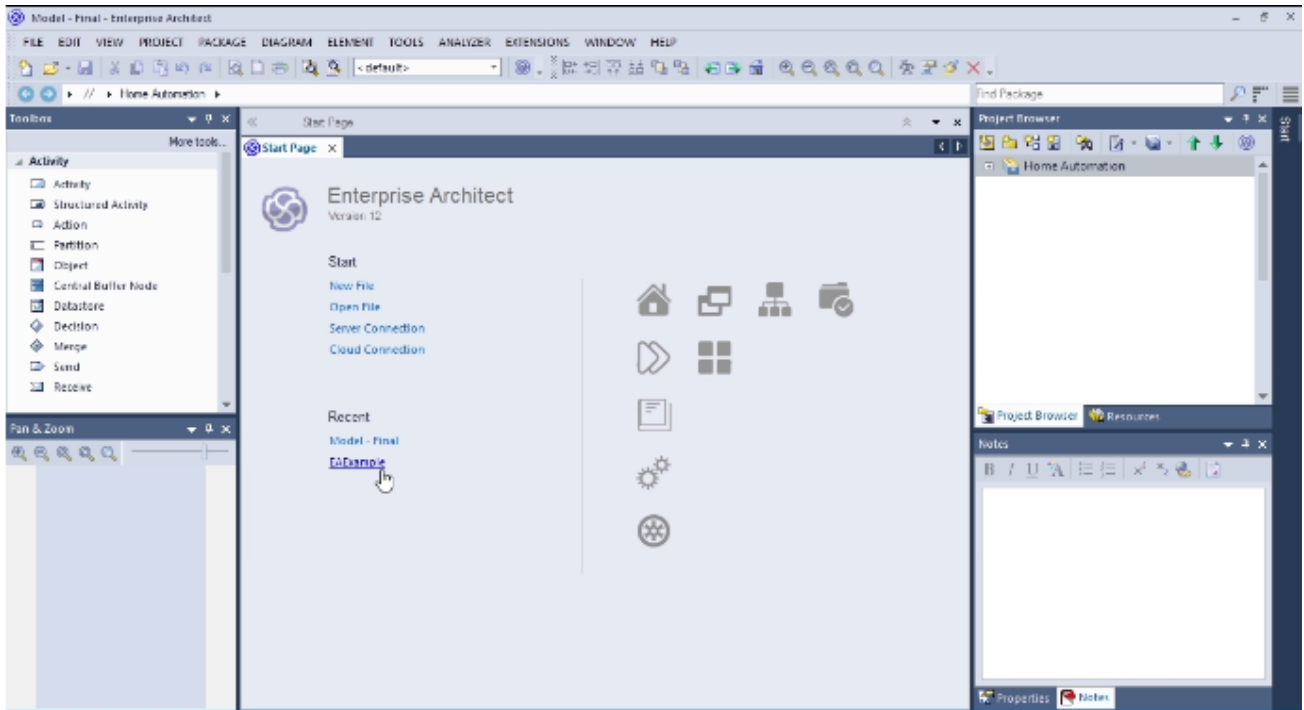
(0:52) The view that is displayed is different from the one in the course. This is partly cosmetic. Sparx Systems has changed how the contents of a model are displayed. Instead of using a Package Diagram, they are now using a Custom Diagram here, which allows them to make the presentation more interesting. If you click on the **Basic Modelling** icon, it will take you to a diagram (shown below) that shows the different UML models grouped under the headings: **Structural Diagrams**, **Behavioral Diagrams** and **Specialized Diagrams**. This is equivalent to the diagram shown in the course materials.



If you open a diagram in a tab and click on a link to open another diagram in the same tab, you can now work back through the diagrams using the small arrow icon at the right-hand end of the tab.



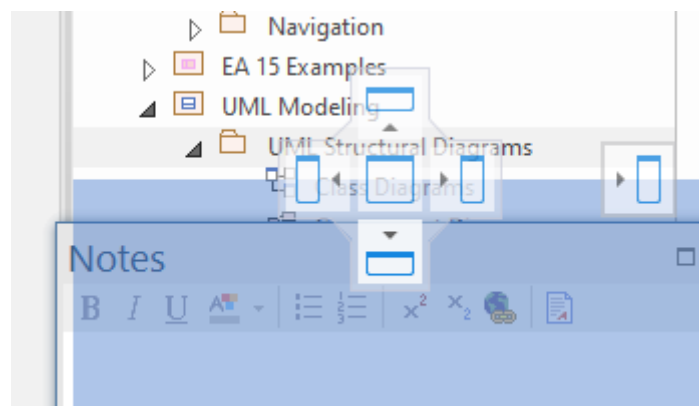
(1:20) The layout in the video at this point looks like this:



On the left are the **Toolbox** and **Pan & Zoom** panes, and on the right are the **Project Browser** and **Notes** panes, with the **Diagram** pane in the middle.

In Version 15, there is an option in the **Start** ribbon under **Visual Style** that opens up a dialogue box in which you can select whether the **Workspace Orientation** is **Right** or **Left**. Having the **Project Browser** on the right is part of the **Right Workspace Orientation**. If you change the orientation in the dialogue, then select a workspace layout by clicking on the **Workspaces** icon and selecting the **Core Modelling** workspace layout, the **Project Browser** pane will move to the right. To get the same layout as I've used at this point in the video, you need to make some other changes. (Note that you don't have to use the same layout as in the video. That layout reflects a default layout when the course was created.)

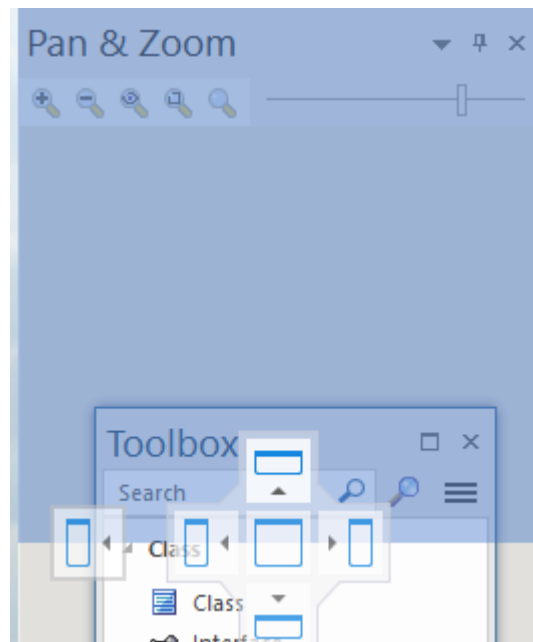
The **Notes** pane shares a space with the **Properties** pane. There are tab selectors at the bottom of the panes. You can select **Notes** rather than **Properties**, and it will display the **Notes** instead of the **Properties**. You can then drag the pane from where it is at the top left over to the lower half of the righthand area where the **Project Browser** pane is. Note how, when you drag a pane to an area, icons appear that allow you to drop that pane into a particular position, as shown in the following diagram.



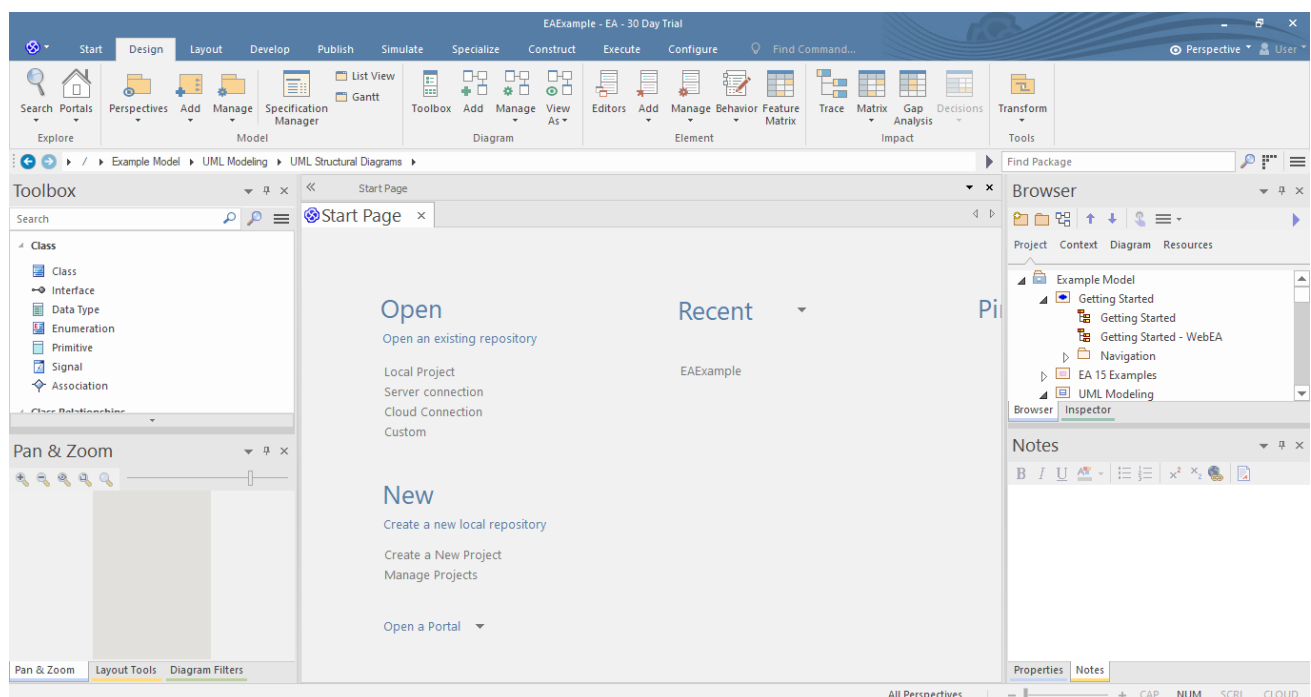
To drag, you position the cursor in the area at the top of the pane where its name is displayed, hold down the left mouse key and move the cursor to the position where you want the pane to go and release the

mouse key. Note that the cursor isn't shown in the screenshot above, but it needs to be over the particular icon that indicates whether the pane is going above, below, to the left or to the right of the pane that is already in that area.

To enable the **Toolbox** pane, you can click the **Toolbox** icon in the **Design** ribbon. This will display it to the right of where the **Pan & Zoom** pane now is on the left of the window. Again, you can drag it to position it at the top of the left-hand area, using those same icons that appear when you drag into that area of the window.

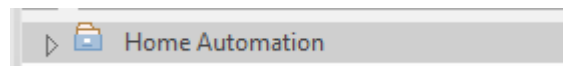


The resulting layout looks like the following diagram.

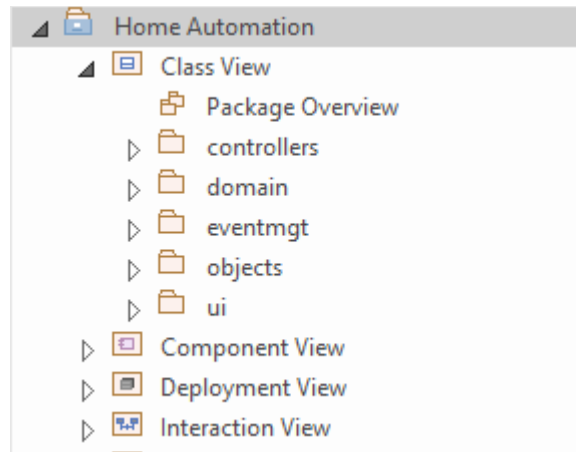


Note that the **View** menu referred to in the video has gone from the user interface. You can click on the **Design** icon in the **Start** ribbon to access a list of panes that can be displayed by clicking on their name.

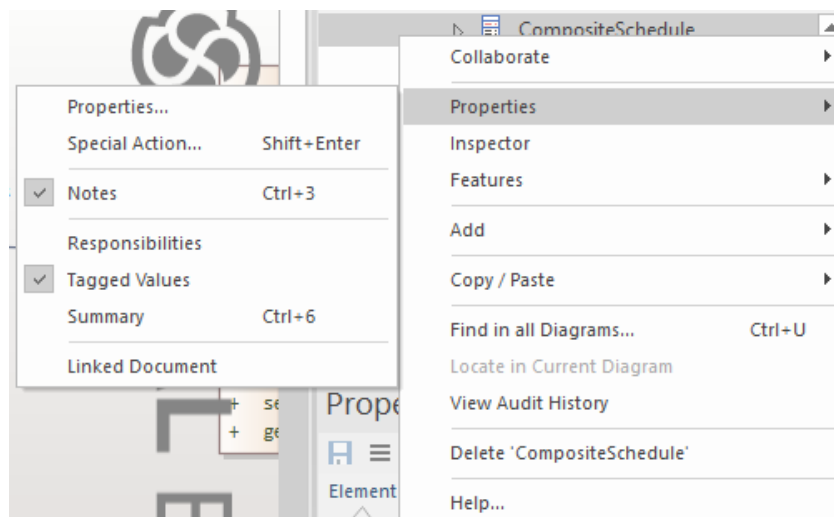
(2:08) The video demonstrates some features of the user interface using the Home Automation model. One of the first differences is that in the **Project Browser**, the plus sign to open up the tree structure of the model has been replaced by a small right-pointing arrowhead like this, which you can click on.



The view of the elements of the model that it displays is essentially the same as in the video.

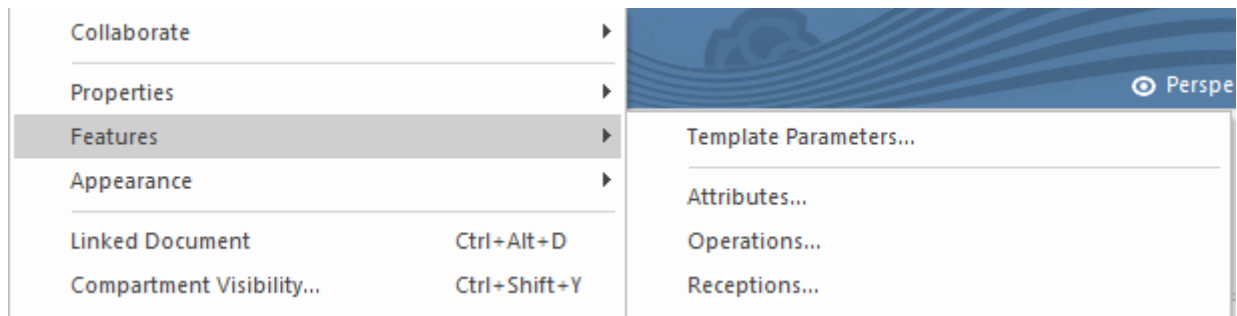


(4:31) The video explains how to access the **Properties** dialogue for elements in a diagram (or for the diagram itself). It is still possible to double-click on an element in the diagram or the background, but you can no longer double-click on an element in the **Project Browser** to open the **Properties** dialogue. However, you can right-click on an element in the **Project Browser** and then select **Properties** from the context menu and **Properties** again from the submenu. (Note that you now have to select it from the submenu.)



(4:50) There is no longer an **Elements** menu item in the menu. To access dialogues that allow you to change aspects of an element, you can right-click on the element and select from the context menu. Alternatively, in the **Design** ribbon, you can select the **Editors** icon, which displays a list of editor panes that you can use to edit the values associated with the currently selected element of the model.

(5:05) **Attributes** and **Operations** are now accessed from the **Features** menu item in the context menu rather than **Features & Properties**.



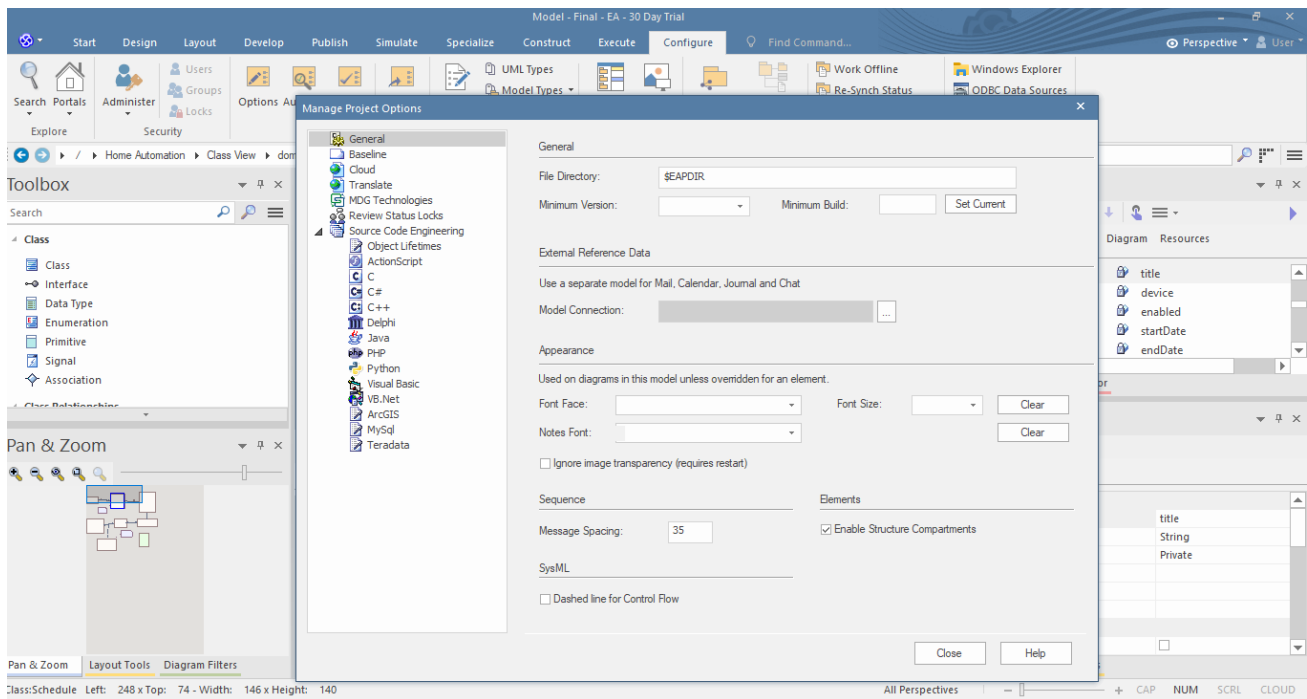
Rather than opening a dialogue box, this now opens a pane, which will appear at the bottom of the **Diagram** pane by default. You may want to keep this open and decide that it is more useful than the **Pan & Zoom** pane (as an example), so you could close the **Pan & Zoom** pane and drag the **Features** pane into its place. Alternatively, you can right-click in the area at the top of the pane where its name is and select **Floating** rather than **Docking**, in which case it will behave like a dialogue box and can be moved around the window and closed and opened as required. If you are using a computer with more than one screen, you can move floating panes into a different screen from the one with the main window.

Features like attributes and operations can be edited in situ in the list in the pane; new ones can be added using the line marked **New Attribute** or **New Operation** at the bottom of the list.

Features					
Attributes Operations Receptions Parts / Properties Interaction Points					
Name	Type	Scope	Stereotype	Alias	Initial Value
title	String	Private			
device	Device	Private			
enabled	boolean	Private			false
startDate	Date	Private			
endDate	Date	Private			
duration	int	Private			
events	TimedEvent	Private			
numberOfEvents	int	Private			
New Attribute...					

If you select an attribute or operation in the list, and the **Properties** pane is open, you will see the properties for that element appear, and they can be edited in the pane. If you select an attribute or operation in the list or in a diagram and then right-click on it, you can select **Properties...** from the context menu to open a dialogue box to edit that particular attribute or operation. If you are using a tool other than EA, it may offer some or all of these navigational routes and possibly others.

(5:20) The overall project settings for EA can now be accessed from the **Configure** ribbon and the **Options** icon.



(5:38) You can close the project that is open with the **Close Project** menu item in the **File** menu. To create a new project, you can click on **Create a New Project** in the **Start Page**, which replaces **New File**. Alternatively, as in version 12, you can select **File > New Project...** to get the file dialogue. The default file type is now **.eapx** rather than **.eap**. As in the video, you can choose a location to put your new project. EA no longer gives you the chance to populate your model with boilerplate content at this point, though you can still add content based on patterns through a wizard later if you want to.

2.3 Common Features in UML

No updates.

3. Class Diagrams

3.1 Purpose of Class Diagrams

No updates.

3.2 Distinction between Objects and Classes

No updates.

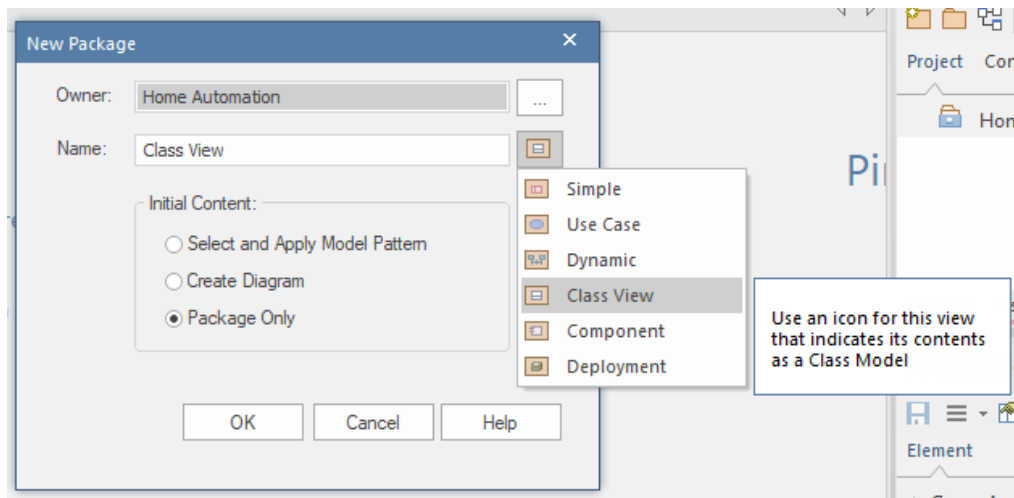
3.3 Basic Notation of Class Diagrams

(2:10) If you created the Home Automation project at the end of the video called *Introduction to the Modelling Software*, then you can just open that project at this point, and it should appear in the **Start Page** and in the **Manage Projects** dialogue box.

If you didn't create the project earlier, you will need to do so now. The default file type is now **.eapx** rather than **.eap**, and you can create a project with the default **.eapx** extension. As in the video, you can choose a location to put your new project. EA no longer gives you the chance to populate your model with boilerplate content at this point, though you can still add content based on patterns through a wizard later if you want to.

(2:50) There is no longer a context menu option to rename the model. To rename it, select **Model** in the **Project Browser** and press the function key **F2**. Note that depending on your computer, you may have to press the **fn** key as well. You can then type the new name in place of the old and hit **Return** or **Enter**.

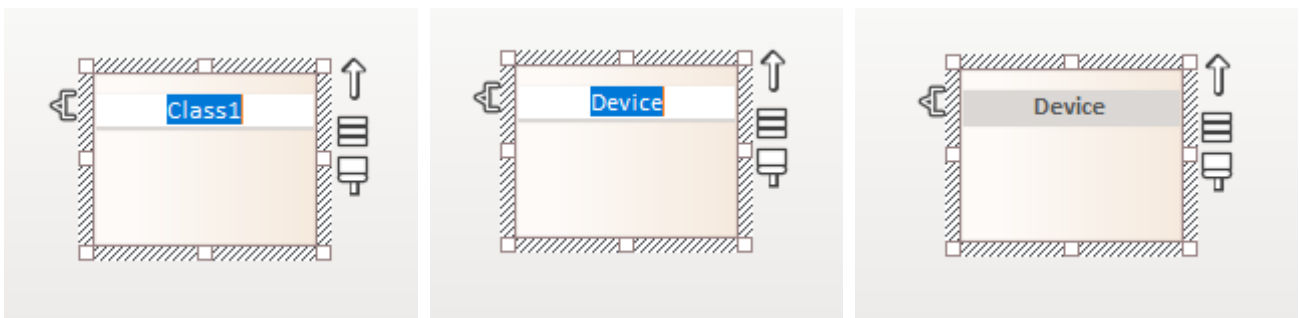
(3:20) To add a view, you can right-click on the model and select **Add View...** The dialogue box that appears is different.



Change the name to **Class View** and also select **Class View** from the drop-down that appears when you click on the button to the right of the **Name** textbox. Change the radio button to select **Package Only**.

(3:51) This is essentially the same as in the video, though the dialogues may look slightly different.

(6:09) EA no longer opens a dialogue box to name the class: it displays the class with the name highlighted in the diagram, and what you type in will replace the machine-generated name (**Class1** in this case).



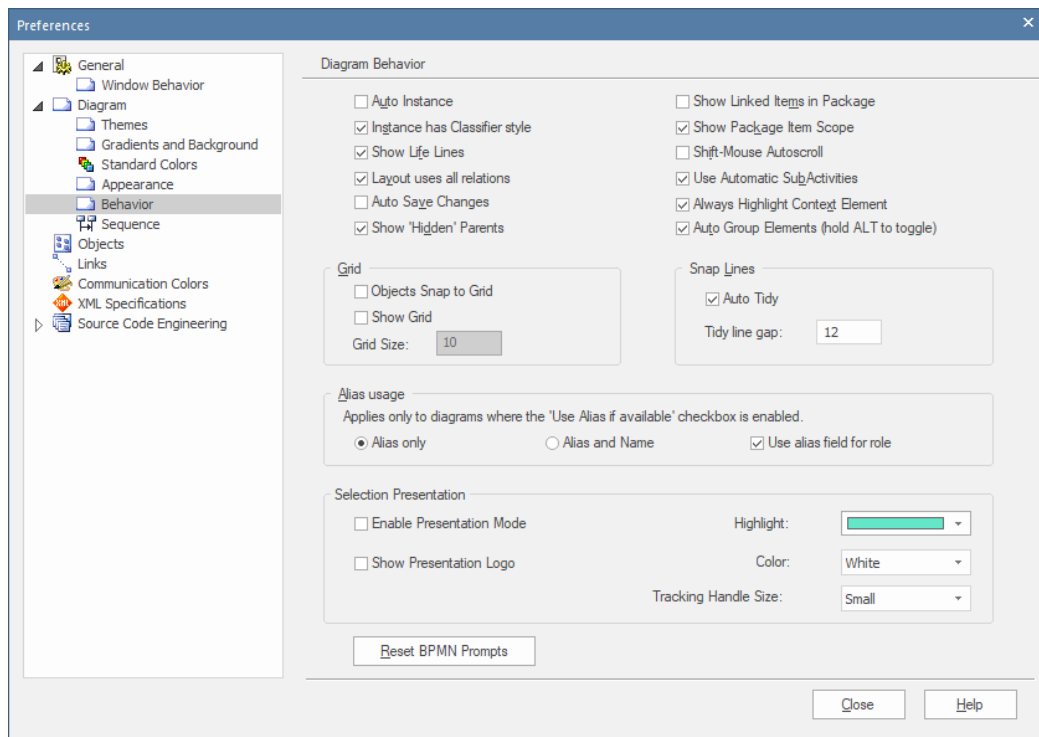
For users of earlier versions that opened the dialogue box, if you were just adding a new class and setting its name, what you type in is essentially the same: the name replaces the name in the class or in the dialogue box, and hitting Return takes the name out of edit mode or closes the dialogue box. However, if you want to add anything else, such as documentation in the **Notes** field, you need to open the **Properties** dialogue box or use the **Properties** or **Notes** pane.

You can access the Properties dialogue box by double-clicking on the name of the class or right-clicking on it and selecting **View Properties...** If the name of the class is selected but not highlighted for editing, you can hit Return, and the **Properties** dialogue will also open.

You can edit the name in situ by selecting it and pressing the function key **F2** or right-clicking and selecting **Edit Selected** from the context menu.

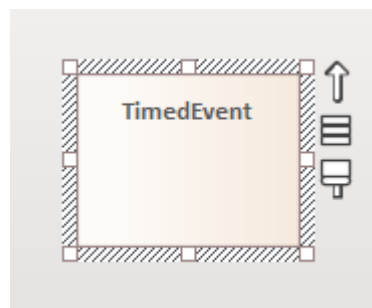
(7:34) Adding more classes will require you to do the same as above, adding the class, editing its name and then adding the Notes through the **Properties** dialogue or the **Notes** pane.

(8:00) As we are adding elements to the diagram, you may want to set the diagram to auto save as you make changes to it. You can do this in the **Preferences** dialogue, which you access from the **Start** ribbon **Preferences** icon. Select **Behavior** in the list on the left, then check the checkbox for **Auto Save Changes**.



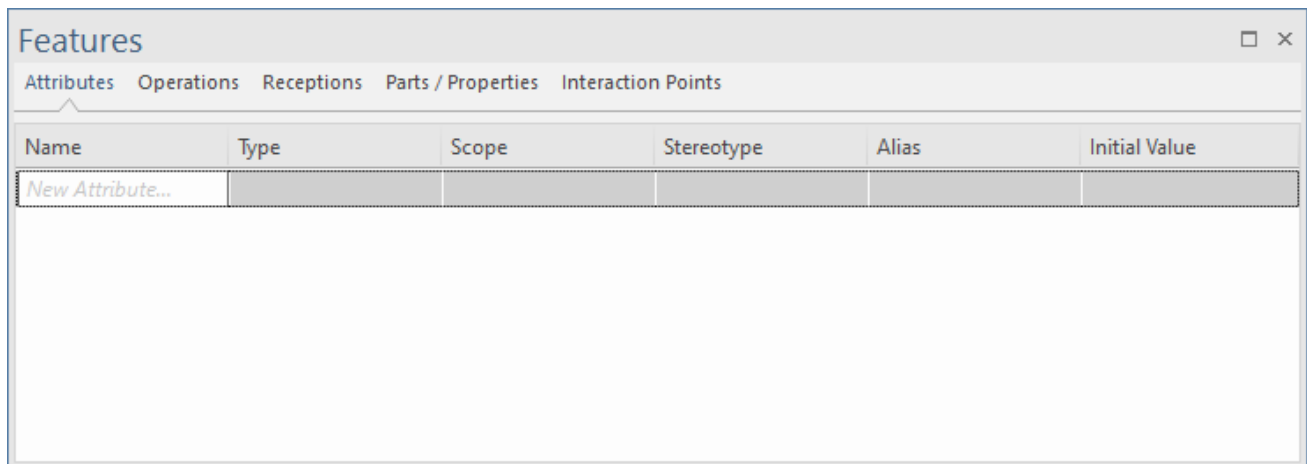
If you don't do this, you will notice an asterisk alongside the name of the diagram in the tab at the top of the **Drawing Pane**. This indicates that it hasn't been saved. You can right-click on the tab and select **Save Changes to 'domain'**. The **Save** button for the diagram is in the **Layout** ribbon.

(10:17) The icons that appear when you select a class have changed in appearance, but the top one is still an arrow and provides the functionality for linking classes together.



3.4 Attributes and Operations - Part 1

(4:29) As we saw earlier, there is no longer a **Features & Properties** context menu item. You right-click on the class and select **Features** and then **Attributes** or **Operations** from the menu. To start with, we're going to add attributes, and we'll use the **Features** pane to do that.

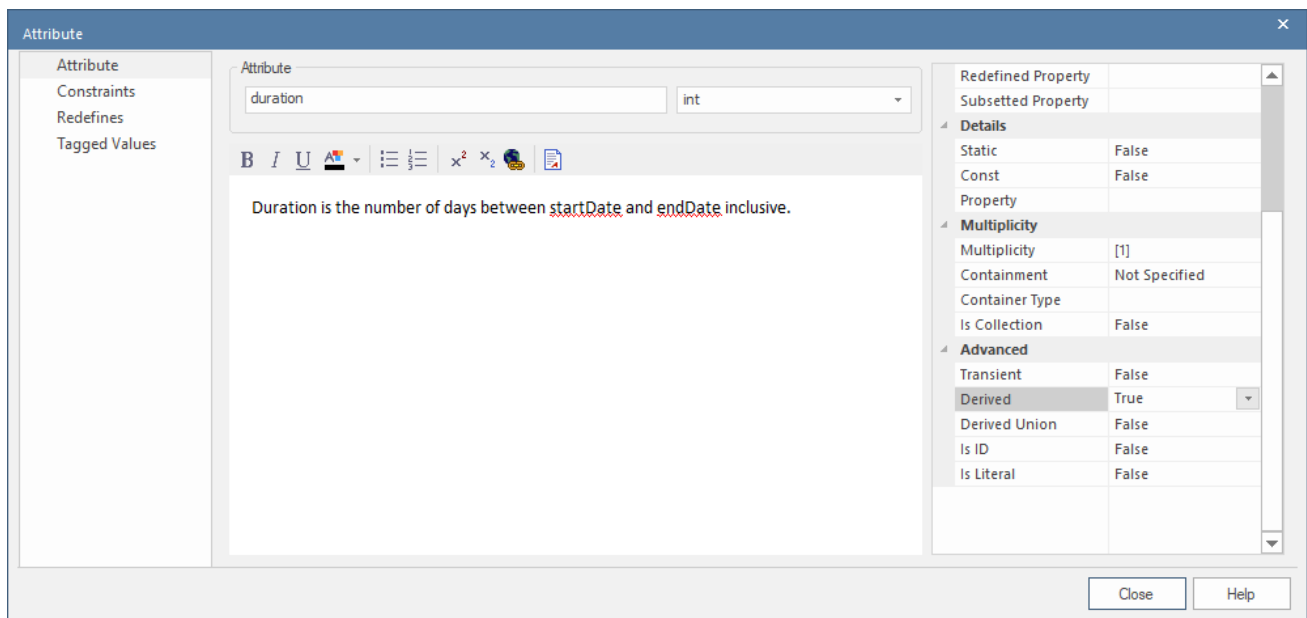


The behaviour of the table of attributes in this pane and the way you add attributes is the same as in the dialogue box in earlier versions. It just doesn't have the other sections of the dialogue for other properties in the same way. You can switch between adding attributes and operations using the tabs at the top of the pane. The **Properties** pane will show the properties for the selected attribute or operation.

(7:28) Double-clicking on the attribute opens the **Properties** dialogue for that attribute rather than the **Features** pane. The same change—to add the initial value of **true** to the **enabled** attribute—can be made in either of these or in the **Properties** pane. The dialogue box no longer limits you to the values true and false for a boolean.

3.5 Attributes and Operations - Part 2

(1:30) Although the dialogue box isn't quite the same, you can open it and use it to set the text into the note and set **Derived** to **True**. Alternatively, you could do this through the **Notes** and **Properties** panes.



(3:00) As before, the reference to the plus sign in the **Project Browser** now means that you should use the small arrow next to the name of the element that you want to expand.

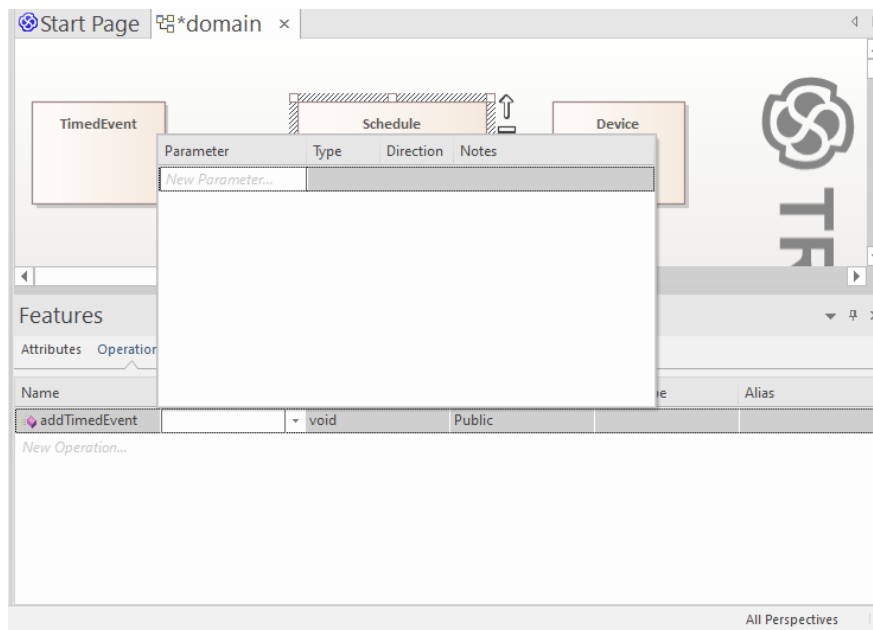
To rearrange the order of the attributes, you can use the **Features** pane and move the attributes in the same way as described in the video using the **ctrl** key and the arrow keys.

The dialogue box to set whether features are sorted alphabetically is now in **Start > Preferences > Preferences... > Objects**.

(4:31) As we saw earlier, there is no longer a **Features & Properties** context menu item. You right-click on the class and select **Features** and then **Attributes** or **Operations** from the menu. This time we're going to add operations, and we'll use the **Features** pane to do that in the same way as we added attributes earlier.

When you add a new operation, the default value for the return type is now **void**.

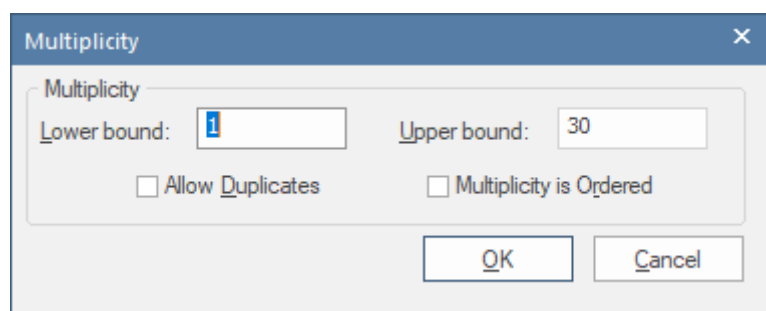
To add a list of parameters from the **Features** pane, you can click on the small arrow to the right of the **Parameters** text box for that operation, and a further dialogue pops up to let you enter parameters.



Note that when you add the second operation **getTimedEvent(index: int) TimedEvent**, you can no longer type the name and type of the parameters into the row in the list in the **Features** pane, but have to use the pop-up window.

3.6 Associations

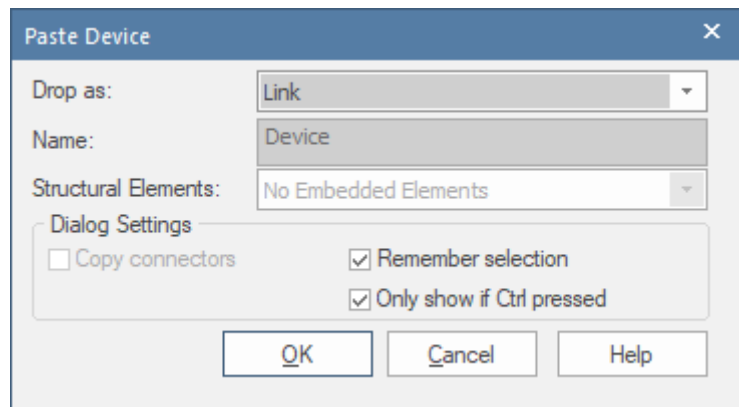
(6:35) To change the multiplicity of the attribute **events**, the simplest way is to edit the text of the attribute in situ. Alternatively, you can use the **Properties** pane and the **Multiplicity** property, which opens the same small dialogue box if you click on the three dots at the right-hand end of the textbox.



3.7 Composition And Aggregation

(5:12) When we drag an element in from the **Project Browser**, the **Paste** dialogue may or may not appear depending on settings. If it doesn't and you want to see it, you can hold down the **Ctrl** key as you drag the

Device into the diagram. You will see that in the dialogue there is a checkbox for **Only show if Ctrl is pressed**. You can check or uncheck this depending on how you want it to work.

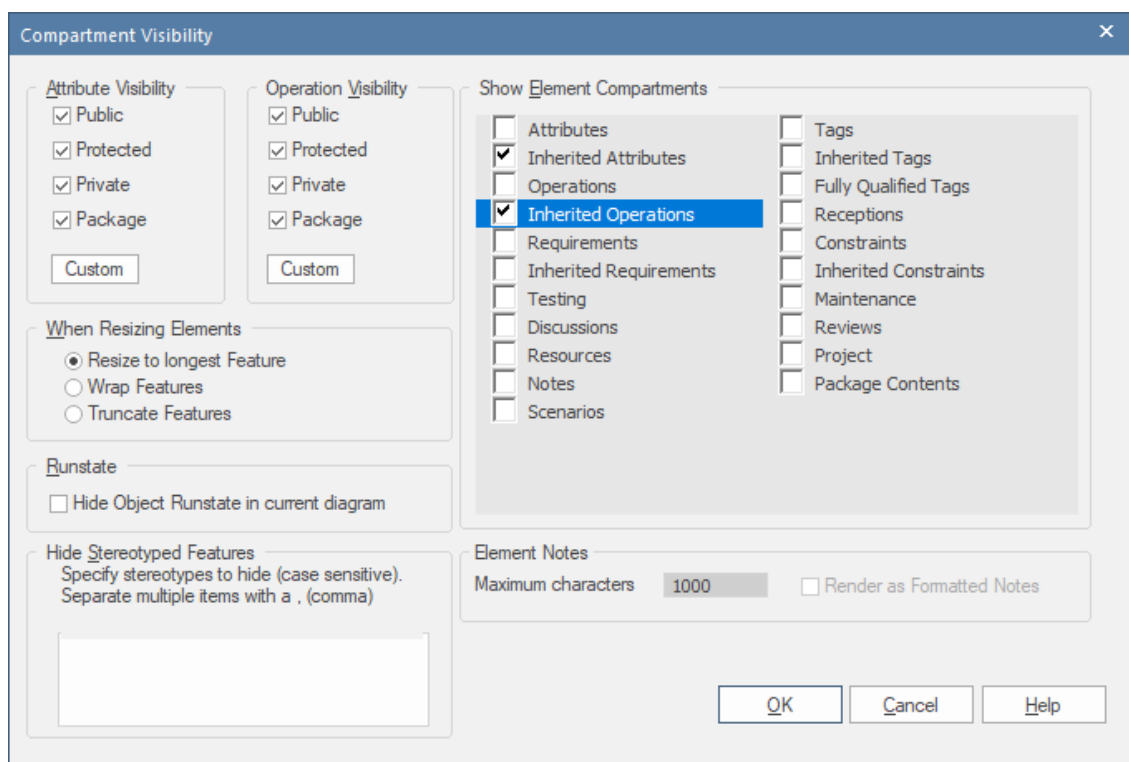


(5:50) There isn't an **Add new element or connector** option in the context menu now. To add a new element, in this case a class, I can right-click in the diagram pane while holding down the **Ctrl** key. This goes straight to a list of the elements that I can add to this diagram. **Class** is at the top of the list.

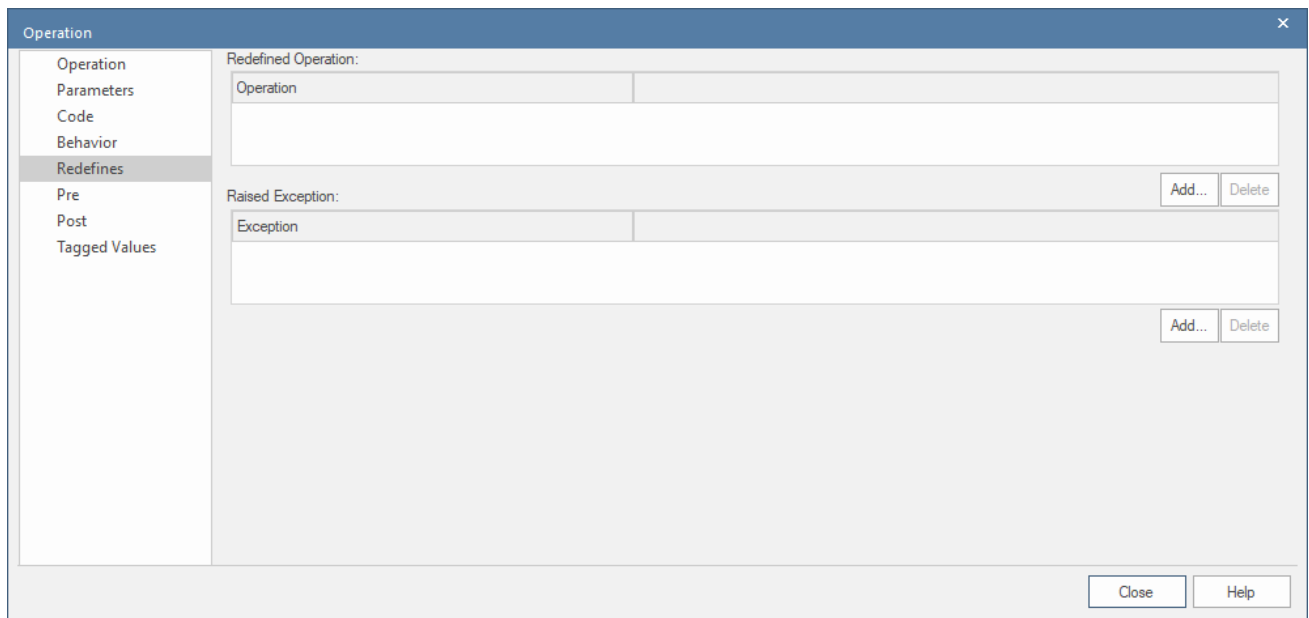
(6:25) The menu that appears when you drag to the **Device** class is different now, but **Aggregation to Part** is in the menu.

3.8 Generalization and Specialization

(4:28) To access the dialogue box to set the visibility of the compartments, you now right-click on the class and select **Compartment Visibility...** from the context menu. The layout of the dialogue has changed, but you still need to check the checkboxes for **Inherited Attributes** and **Inherited Operations**.



(5:35) The **Operation** dialogue box has changed, and **Redefines** is now selected from the list of categories on the left rather than being a tab in the area at the bottom right.



4. Object Diagrams

4.1 Purpose of Object Diagrams

No updates.

4.2 Basic Notation of Object Diagrams

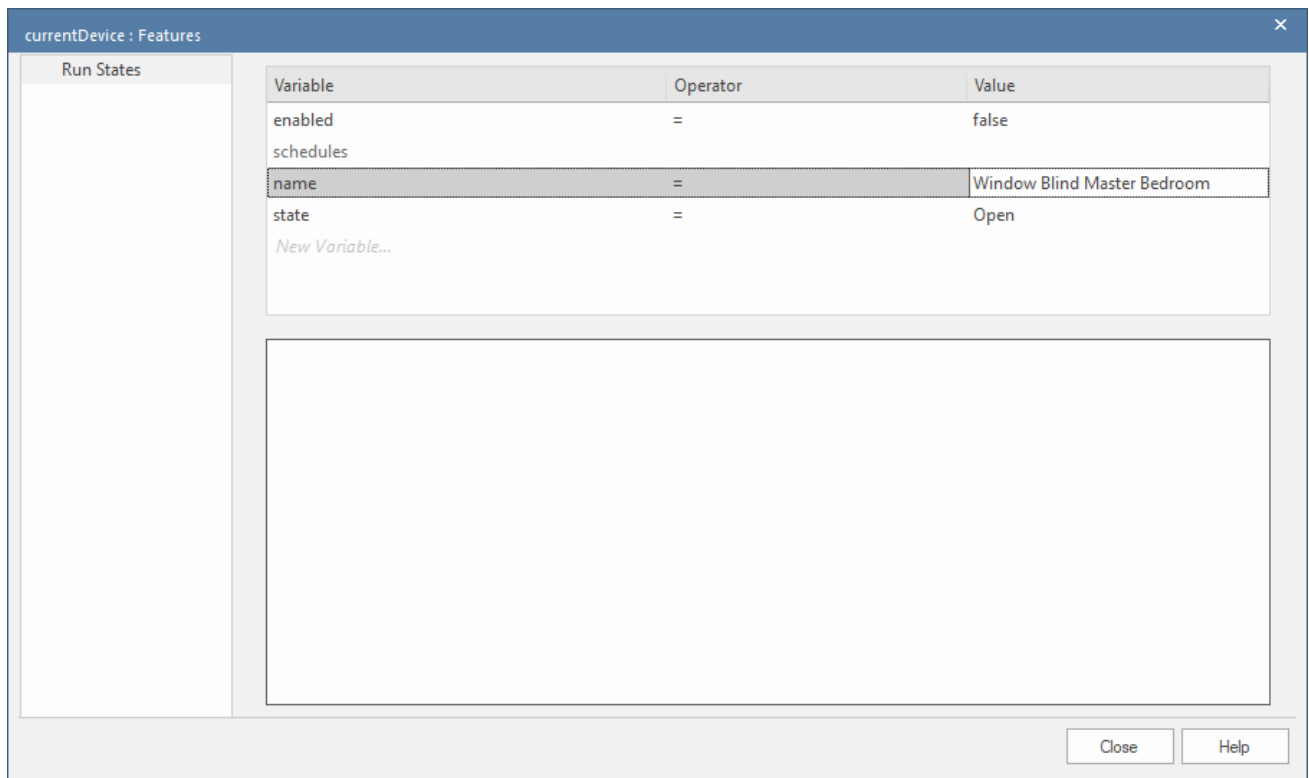
Most of this video shows actions in the user interface that we have already seen in earlier videos.

(4:50) The option to add a **Directed Link** no longer exists in the **Quick Linker** menu, so you have to add just an **Association** and then add the navigability information to it or use the same options as described for the other link, using the right-click context menu and selecting **Advanced > Change Direction > Source -> Destination**.

Although EA now uses **Association** here, technically, these are Links according to the UML specification. A link is an instance of an association in the same way that an object is an instance of a class.

4.3 Object State

(3:00) As before, there is no longer a **Features & Properties** menu item. You can access **Set Run State...** through the **Features** menu item. The dialogue has changed and now displays a list of attributes for which slot values can be set. (Slots are instances of attributes.)



(5:15) Adding attributes via the **Project Browser** now requires you to right-click on the element (in this case **TimedEvent**) and select **Features** and then **Attributes** from the submenu. It opens the **Features** pane rather than the **Properties** dialogue.

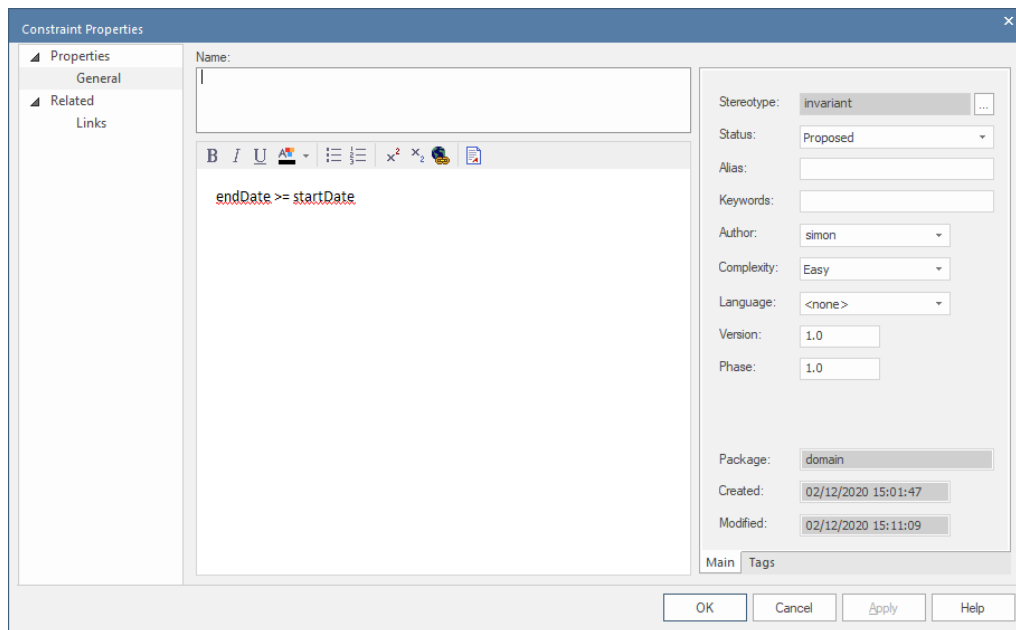
5. Advanced Class Diagrams

5.1 Advanced Class Diagrams

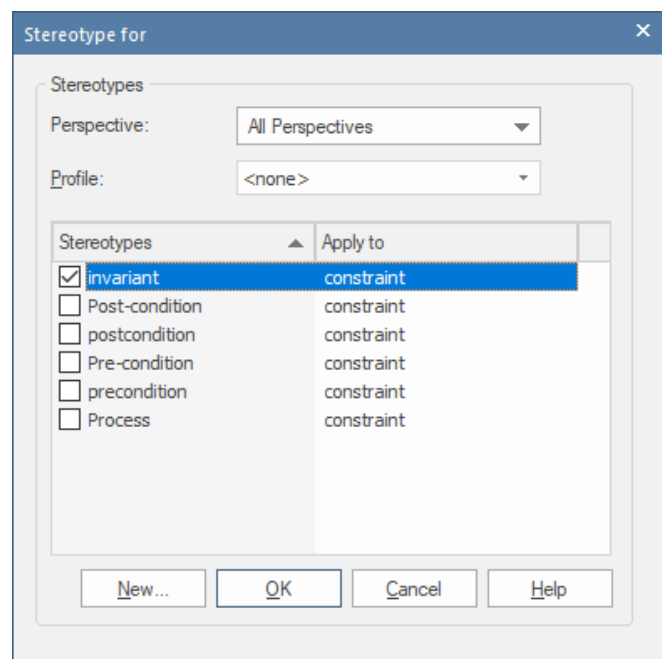
No updates.

5.2 Advanced Associations

(4:50) Adding the **Constraint Note** no longer automatically opens the dialogue box. If you double-click on the **Constraint Note**, it will open the **Properties** dialogue. You can put the constraint text into the notes field.



To set the **invariant** as the stereotype, you can click on the three dots button next to the **Stereotype** field at the top right of the dialogue. In the dialogue that appears, set **Perspective** to **All Perspectives** or **All UML**, and then set **Profile** to **<none>**. You can select **invariant** from the list that is displayed.



(7:05) When you add the association class, you will have to double-click on the class to open the dialogue or edit it in situ to change the name.

(7:37) Using the **Features** pane, you won't be able to add the multiplicities, but you can add the attributes and then use the **Properties** dialogue or pane to change the **Multiplicity** values.

5.3 Interfaces And Enumerations

In Java, interfaces are also now used by the functional features introduced in Java 9, in particular the use of functional interfaces in lambdas. Also, in Java operations of interfaces can now have default implementations, so that they are more like abstract classes.

In Java, enumerations (the enum type) can also have attribute values together with constructors and methods to get the values.

These are language-specific issues and not about UML, but affect how you might choose to model interfaces and enumerations if you are developing the model for a Java implementation.

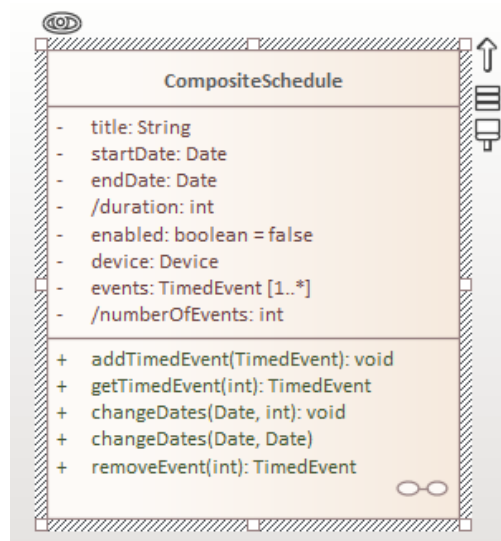
5.4 Internal Structure Of Diagrams

(2:37) There is no longer an Edit menu, so the quickest way to copy an element from a diagram is to select it and press **Ctrl-C** or right-click and select **Copy / Paste** and then **Copy**.

To paste the new class in, you now need to right-click on the diagram and select **Paste** then **Element(s) as New**.

To add the composite structure, the process has changed more significantly, as there is no longer an option in the **Compartment Visibility** dialogue to switch on the compartment.

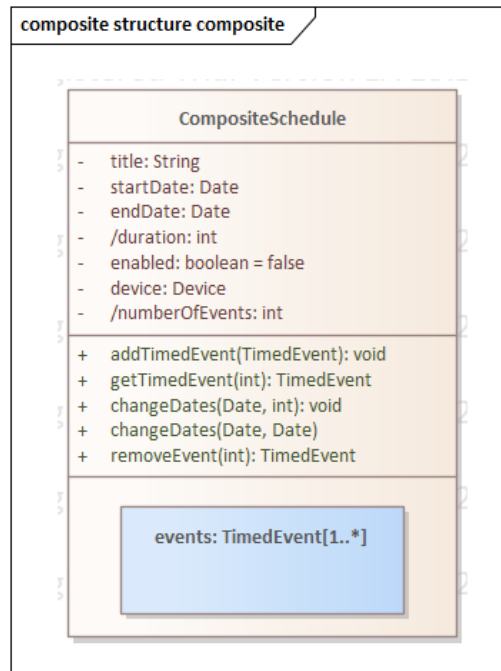
The first step is to right-click on the element (in this case **CompositeSchedule**) and select **New Child Diagram** and then **Composite Structure Diagram**. This will add a small icon to outside the top-left corner of the element in the diagram and another inside the bottom right-hand corner.



The second step is to open the diagram and add the new property based on **TimedEvent**. If you double-click on **CompositeSchedule**, this will take you into the new diagram, which has the same name as the class. (Note that this changes the previous behaviour where double-clicking opened the **Properties** dialogue for the class.) You can now drag **TimedEvent** into the diagram. When you do, select **Property** as the type in the dialogue. You can rename the property that has been added to the diagram as **events**. Now click on the back arrow in the diagram tab to go back to the parent diagram: the one called **composite**.

Finally, go back and right-click on **Composite Schedule** again and select **New Child Diagram** then **Show Composite Diagram in Compartment**. Note that you may have to save the child diagram for the change of name to be reflected in **CompositeSchedule**.

If you dragged **TimedEvent** in as a property, the type will already be showing alongside the name of the property as **events: TimedEvent**. To change the multiplicity, you have to drill down into the child diagram by double-clicking on **CompositeSchedule**. You can then select the **events** property element in the diagram and set **LowerBound** and **UpperBound** in the **Properties** pane. You can also set the **Ordered** checkbox. Then you can return to the parent diagram.



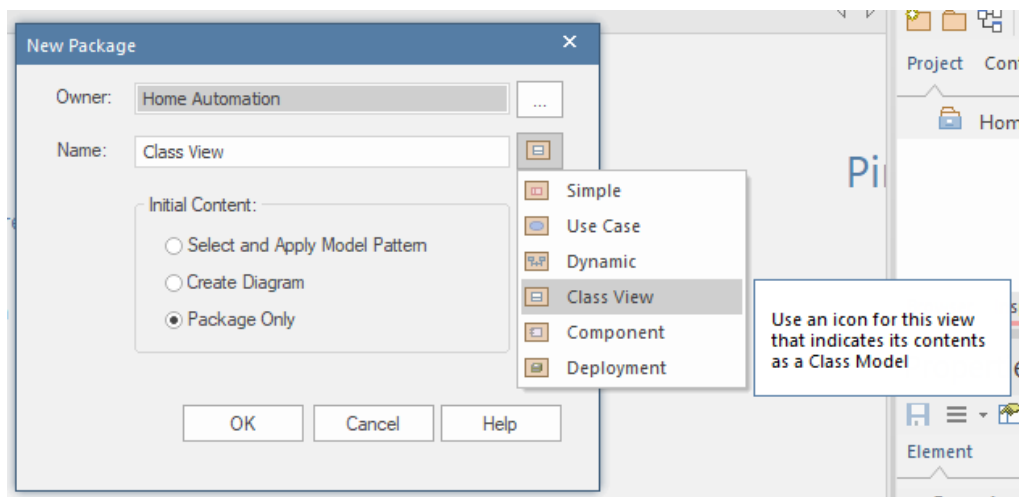
6. Use Case Diagrams

6.1 Purpose Of Use Case Diagrams

No updates.

6.2 Basic Notation Of Use Case Diagrams - Actors And Use Cases

(3:40) We added a new view earlier to create the **Class View**. Here's a screenshot of what we did. You need to do something similar to create the **Use Case View**.



To add a view, you can right-click on the model and select **Add View...**. The dialogue box that appears is different.

Change the name to **Use Case View** and also select **Use Case View** from the drop-down that appears when you click on the button to the right of the **Name** textbox. Change the radio button to select **Package Only**.

(5:00) As with the addition of classes to a class diagram, adding the boundary now doesn't automatically open the **Properties** dialogue box, but we can type the name directly into the **Boundary** that it creates in the diagram.

(6:25) As before, adding a use case to the diagram doesn't open the **Properties** dialogue, but you can type the name in directly to the **Use Case**. You can document it by typing the note into the **Notes** pane without needing to open the dialogue box.

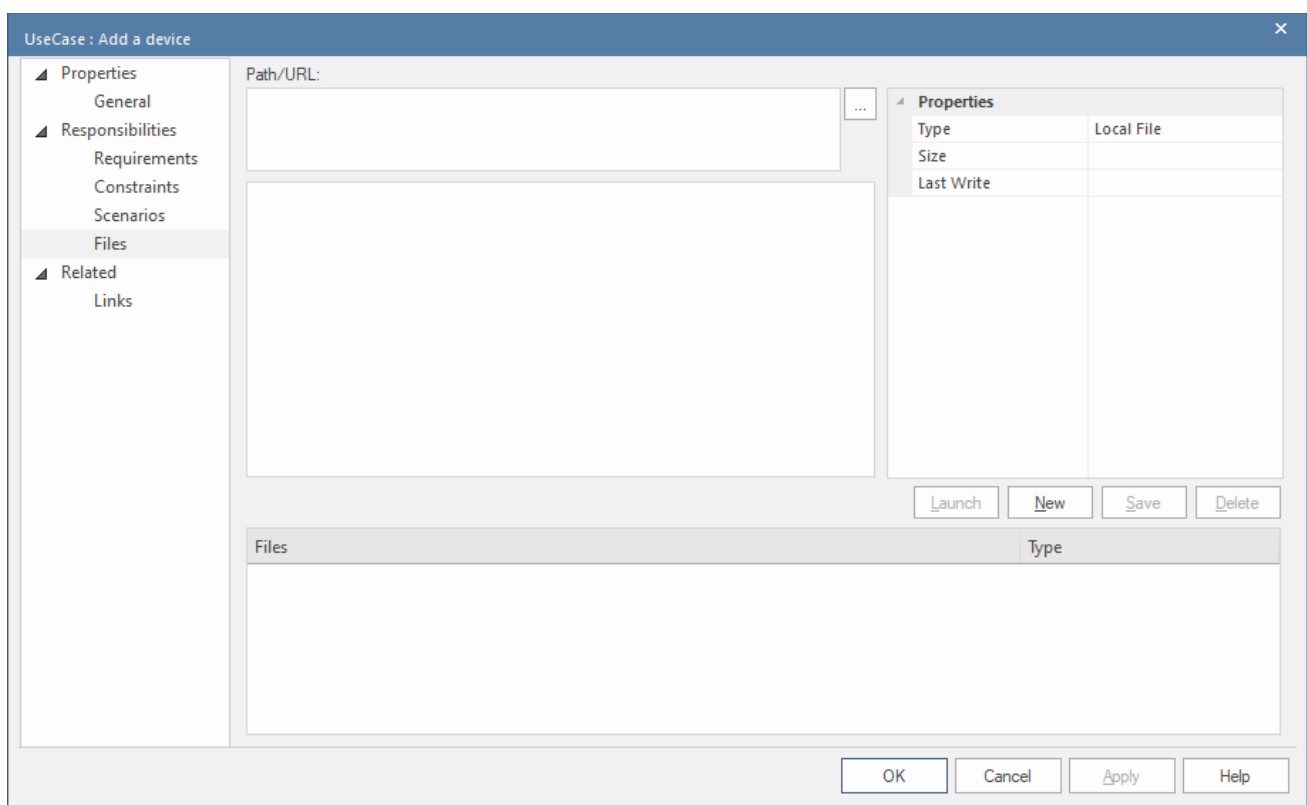
The same applies when you add an **Actor**.

6.3 Relationships In Use Case Diagrams

(8:30) When you link the «**extend**» relationship to the extension point, a note symbol is automatically added to the diagram, though it may not be displayed next to the relationship, so you may need to move it. We want a constraint rather than a note, so you can delete the note and then follow the instructions to add a constraint. The dialogue box for setting the constraint itself is the same as in section 5.2 above.

6.4 Specifying Use Cases

(3:40) The dialogue for adding files to a model element is slightly different and is shown below.



(4:40) To view the **Resources**, you now select the **Resources** tab at the top of the **Project Browser**. Rather than **Linked Document Templates** and then **Model**, you select **Linked Documents** and then **Model Templates**. The option to save is in the **Layout** ribbon, or you can right-click on the tab at the top of the pane.

(5:40) The menu item in the right-click context menu is now **Linked Document** rather than **Add Linked Document**. The Use Case Template that was created is now in a section labelled **Custom Templates** rather than **User Templates**.

(6:50) There is no longer a **Structured Specification** tab. The Scenario Editor is now accessed using the **Structure Editor** button in the **Scenarios** section of the **Properties** dialogue, which opens it up as a pane in the same area as the diagrams, as shown below.

UseCase : Add a device

Start Page
Configuration
Scenarios

Type: Basic Path
Scenario: Basic Path

Step	Action	Uses	Results	State
1	The user selects Add Device from the menu			
2	The system presents the screen allowing the user to add a device			

Entry Points
Context References
Constraints

Step	Path Name	Type	Join
0	Basic Path	Basic Path	-

From this point on, we hope that you have a good enough idea of how the version 15 of Enterprise Architect differs from earlier versions to proceed with the rest of the videos.